

HDFC Mutual Fund Purchase Optimizer - API Documentation (v1.0)

Overview

This document serves as the canonical reference for the HDFC Mutual Fund Purchase Optimizer backend API (v1). The API follows RESTful principles and uses JSON for request and response payloads.

Base URL: `http://localhost:8000/api/v1` **Authentication:** Bearer Token (JWT) **Content-Type:** `application/json`

1. Authentication & Security

All protected endpoints require a valid JWT token in the `Authorization` header.

Header Format: `Authorization: Bearer <access_token>`

Common Response Codes

Code	Description
200	Success
400	Bad Request (Validation Error)
401	Unauthorized (Missing/Invalid Token)
403	Forbidden (Insufficient Permissions)
404	Not Found
500	Internal Server Error

2. Standard Response Wrap

The API endeavors to return consistent JSON structures.

Success Response:

```
{
  "data": { ... }, // Payload
  "message": "Operation successful",
  "status": "success"
}
```

Error Response:

```
{
  "detail": "Error description here",
}
```

```
"code": "ERROR_CODE"
}
```

3. Core API Endpoints

3.1 Authentication

Method	Endpoint	Description	Auth Required
POST	/auth/signup	Register a new user	No
POST	/auth/login	Obtain access token	No
GET	/auth/me	Get current user profile	Yes

POST /auth/signup Request Body:

```
{
  "email": "user@example.com",
  "password": "securepassword123",
  "name": "John Doe"
}
```

POST /auth/login Request Body: application/x-www-form-urlencoded
- username: Email address - password: User password

Response:

```
{
  "access_token": "eyJhbGciOiJIUzI1Ni...",
  "token_type": "bearer",
  "user": { "id": 1, "email": "user@example.com", "role": "user" }
}
```

3.2 Funds & Market Data

Method	Endpoint	Description	Auth Required
GET	/funds	Get list of funds (Home Page)	No
GET	/fund/{scheme_code}	Get fund details	No
GET	/funds/search?query=...	Search funds	No
GET	/market/indices	Get market indices	No

GET /funds **Query Parameters:** - **limit:** Number of funds (default: 10) - **offset:** Pagination offset (default: 0) - **category:** Filter by category (optional)

Response:

```
[
  {
    "id": "100475",
    "name": "HDFC Top 100 Fund",
    "nav": 854.32,
    "metrics": { "cagr_3y": 0.12, "sharpe": 1.2 }
  },
  ...
]
```

GET /fund/{scheme_code} **Response:**

```
{
  "id": "100475",
  "name": "HDFC Top 100 Fund",
  "nav": 854.32,
  "metrics": { "cagr_3y": 0.12, "sharpe": 1.2 }
}
```

3.3 Portfolio & Investments

Method	Endpoint	Description	Auth Required
GET	/portfolio	View user holdings	Yes
POST	/invest	Purchase mutual funds	Yes
POST	/sip/pause	Pause SIP (Proposed)	Yes

POST /invest **Request Body:**

```
{
  "scheme_code": "100475",
  "amount": 10000.0,
  "type": "lumpsum" // or "sip"
}
```

3.4 Risk Profile (Proposed)

Method	Endpoint	Description	Auth Required
POST	/risk-profile/{user_id}	Calculate Risk Score	Yes
GET	/risk-profile/{user_id}	Get Risk Profile	Yes

POST /risk-profile/{user_id} Request Body:

```
{
  "age": 28,
  "income": 1200000,
  "emergency_fund": "partially",
  "investment_duration": "3-5",
  "market_drop_reaction": "invest_more",
  "return_expectation": "10-14",
  "experience": "some"
}
```

Response:

```
{
  "risk_score": 0.55,
  "risk_profile": "Moderate",
  "details": { ... }
}
```

3.5 Advisor Tools (Proposed)

Method	Endpoint	Description	Auth Required
GET	/advisor/clients	List managed clients	Yes (Advisor)
POST	/advisor/invest/otp	Generate OTP for client transaction	Yes (Advisor)
POST	/advisor/distribute	Execute transaction with OTP	Yes (Advisor)

POST /advisor/invest/otp Request: { "client_id": 1, "fund_id": "100475", "amount": 5000, "action": "invest" } Response: { "message": "OTP sent", "transaction_id": "tx_123" }

POST /advisor/distribute Request: { "transaction_id": "tx_123", "otp": "123456" } Response: { "message": "Investment successful", "units_allocated": 12.5 }

3.6 Admin Tools (Proposed)

Method	Endpoint	Description	Auth Required
GET	/admin/users	List all users	Yes (Admin)
GET	/admin/funds	List all funds	Yes (Admin)
POST	/admin/funds	Add new fund	Yes (Admin)

3.7 User Management (Proposed)

Method	Endpoint	Description	Auth Required
PUT	/users/me	Update user profile	Yes
PUT	/users/password	Change password	Yes
DELETE	/users/me	Deactivate account	Yes

PUT /users/me Request Body:

```
{
  "name": "Updated Name",
  "age": 29,
  "income": 1500000,
  "risk_profile": "Aggressive"
}
```

Response:

```
{
  "id": 1,
  "name": "Updated Name",
  "email": "user@example.com",
  "role": "user",
  "profile": { "age": 29, ... }
}
```

PUT /users/password Request Body:

```
{
  "current_password": "securepassword123",
  "new_password": "newsecurepassword456"
}
```

4. Verification & Testing

Interactive Documentation (Swagger UI)

For **implemented** APIs, you can interact directly using the built-in Swagger UI: 1. Ensure the backend server is running (`uvicorn src.main:app`). 2. Navigate to **`http://localhost:8000/docs`**. 3. Use the “Try it out” button on any endpoint to send live requests.

Testing Proposed APIs

The APIs marked as **(Proposed)** (Risk Profile, Advisor, Admin, OTP) are currently *specified* but not yet *implemented*. To verify these: 1. They will **not** appear in `/docs` yet. 2. We must implement the corresponding routers and logic in the backend first. 3. Once implemented, they will automatically appear in the Swagger UI for verification.
